

EDUCATION

M.Sc. in Computer Engineering, GPA: 3.89/4

2021 - 2023

Sharif University of Technology

Tehran, Iran

- **Thesis Title:** Deep Multi-Object Tracking by Part-Based Re-Identification in Soccer Matches
- **Selected Courses:** Machine Learning - Advanced Machine Learning - Deep Learning - Digital Image Processing - Advanced 3D Computer Vision - Natural Language Processing - Digital Signal Processing

B.Sc. in Computer Engineering, GPA: 3.75/4

2017 - 2021

Isfahan University of Technology

Isfahan, Iran

- **Thesis Title:** Biometric Image Processing for Authentication and Identification of Persons
- **Selected Courses:** Design and Analysis of Algorithms - Artificial Intelligence - Applied Linear Algebra - Foundation of Computer Vision - Multimedia Systems - Computational Intelligence

RESEARCH INTERESTS

Computer Vision

Vision-Language Models

Deep Learning

RESEARCH EXPERIENCE

Research and Develop Position

Oct 2024 - Present

AlgoNet Research

Tehran, Iran

- **Description:** Development of a novel relative positional encoding method for length extrapolation of LLMs in next token prediction task by learning context-aware biases.
- **Description:** Development of a comprehensive text detection and recognition system using multimodal VLMs for Persian document analysis.

Research Assistant

Nov 2021 - Aug 2023

Image Processing Lab, Sharif University of Technology

Tehran, Iran

- **Description:** The proposal and implementation of a multi-purpose part-based person representation method for part-based tracking of persons in team sport videos.
- **Description:** The proposal and implementation of a pair-wise distillation approach by transferring inter-class similarities for semantic segmentation.
- Supervisor: Prof. Shohreh Kasaei

Research Internship

Summer 2022

Image and Signal Processing Lab, UCLouvain

Louvain-la-Neuve, Belgium

- **Description:** The development and implementation of a multi-object tracking solution exploiting re-identification descriptors.
- Supervisor: Prof. Christophe De Vleeschouwer

SELECTED PUBLICATIONS

Full list of papers on Google Scholar

1. **A. M. Mansourian**, A. M. Babaei, and S. Kasaei. “Enriching Knowledge Distillation with Cross-modal Teacher Fusion.” *CVPR, Findings* 2026.
2. **A. M. Mansourian***, M. H. Sameti*, A. Marioriyad, M. H. Rohban, and M. Soleymani. “No Concept Left Behind: Test-Time Optimization for Compositional Text-to-Image Generation.” *ICASSP*, 2026.
3. **A. M. Mansourian**, A. Jalali, R. Ahmadi, and S. Kasaei. “Attention as Geometric Transformation: Revisiting Feature Distillation for Semantic Segmentation.” *WACV, Workshop* 2026.
4. **A. M. Mansourian***, A. Veisi*, and H. Amirzadeh*. Context-aware Biases for lenght Extrapolation. *EMNLP*, 2025.
5. **A. M. Mansourian**, R. Ahmadi, M. Ghafouri, A. M. Babaei, E. B. Golezan, Z. Y. Ghamchi, V. Ramezani, A. Taherian, K. Dinashi, S. Kasaei. “A Comprehensive Survey on Knowledge Distillation.” *Transactions on Machine Learning Research (TMLR)* 2025.
6. V. Somers, V. Joos, A. Cioppa, S. Giacola, S. A. Ghasemzadeh, F. Magera, B. Standaert, **A. M. Mansourian**, X. Zhou, S. Kasaei, and others. “SoccerNet Game State Reconstruction: End-to-End Athlete Tracking and Identification on a Minimap.” *CVPR, Workshop* 2024.
7. **A. M. Mansourian**, V. Somers, C. De Vleeschouwer, and S. Kasaei. “Multi-task Learning for Joint Re-identification, Team Affiliation, and Role Classification for Sports Visual Tracking.” *ACM MM, Workshop* 2023.

HONORS AND AWARDS

- Participating as reviewer at CVSports workshop, CVPR 2024/2025/2026
- Participating as reviewer at WACV and ICCV conferences 2025
- Participating as a part of organizer team of SoccerNet Challenges 2022/2023/2024
- Best paper award at International Conference on Machine Vision and Image Processing Mar 2024
- **Ranked 5th** at National Computer Engineering Olympiad Oct 2021
- **Ranked 10th (top 0.05%)** in the National Entrance Exam for M.Sc. degree in Computer Engineering Sep 2021
- **Ranked 7th** among undergraduate Computer Engineering students 2017 - 2021
- **Ranked in top 1%** in the National Entrance Exam for B.Sc. of Iran, among more than 148,000 students in the field of Mathematics and Physics Jul 2017

TEACHING EXPERIENCE

Teaching Assistant

Sharif University of Technology

Tehran, Iran

- Presenting recent advancements in Computer Vision and Pytorch tutorial Spring 2024/2025
- Digital Image Processing (Supervisor: Prof. Shohreh Kasaei) Spring 2023/2024
- Advanced 3D Computer Vision (Supervisor: Prof. Shohreh Kasaei) Fall 2022/2023
- Machine Learning (Supervisor: Prof. Hamid Beigy) Spring 2023

Teaching Assistant

Isfahan University of Technology

Isfahan, Iran

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| – Design and Analysis of Algorithms (Supervisor: Dr. Mohammad Reza Heidarpour) | Fall 2019, Spring 2023 |
| – Theory of Formal Languages (Supervisor: Dr. Hossein Falsafain) | Spring 2021 |
| – Digital System Design Lab (Supervisor: Dr. Nader Karimi) | Spring 2020 |
| – Basis of C Programming Lab (Supervisor: Dr. Elham Mahmoudzadeh) | Fall 2018, Spring 2019 |

SKILLS

- **Programming Languages:** Python, C, C++, Java, MATLAB, R
- **Frameworks and Libraries:** OpenCV, Numpy, Pytorch, Tensorflow, Keras
- **Technical Skills:** Git, Latex, Docker, HTML, CSS, Microsoft SQL Server
- **Languages:** English, Persian, Kurdish

SELECTED PROJECTS

See full list of projects on github.com/AmirMansurian

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| <ul style="list-style-type: none">• Knowledge Distillation
Implementation of a novel and simple feature distillation approach for semantic segmentation by feature refinement via attention mechanism. (Link) | <ul style="list-style-type: none">• Multi-purpose Person Representation
Training a part-based method for joint re-identification, role classification, and team affiliation in multi-task manner for sports visual tracking. (Link) |
| <ul style="list-style-type: none">• Length Exerapolation of LLMs
Context-aware biases for a relative positional encoding for decoder-based large language models. (Link) | <ul style="list-style-type: none">• Tracking Lab
An easy-to-use modular framework for multi-object pose/segmentation/bbox tracking. (Link) |
| <ul style="list-style-type: none">• Persian Question Answering
An Experimental Survey on Farsi Transformer-based Retrival Question Answering. (Link) | <ul style="list-style-type: none">• Efficient Optical Flow Estimation
Rethinking RAFT model for proposing an efficient optical flow estimation method. (Link) |

TEST SCORES

- **TOEFL iBT:** 5.5 (110) (Reading: 5, Listening: 6, Speaking: 5, Writing: 5.5)